



ABC Private Limited

Travel & Expense Management App

Pre-travel approval · Booking · Expense statement with bills · Approvals · Payment · Analytics ·
PWA

PROJECT SUMMARY DOCUMENT

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1. Executive Summary

This project delivers a complete, ready-to-run **Travel & Expense Management application** for ABC Private Limited. It digitises the full corporate travel cycle – from the sales person's pre-travel approval request, through Business Head and Managing Director approval, booking of flights and hotels by the Travel Assistant as per the Matrix of Authority, to the post-travel expense statement with attached bills, its approval by the Business Head and HR Head, and finally accounting and payment by the Accountant.

The application has a **backend server with a SQLite database** that stores all users, sign-in/sign-up records, requests, approvals, bookings, expense lines, bills and payments. Access is controlled by a **login page (email id + password)** with sign-up; the first user to sign up becomes the Admin, and afterwards only email ids added by the Admin can sign up.

Admins get a **visual analytics dashboard** with employee-wise, destination-wise, airline-wise, department-wise and monthly spend, a lead-time-versus-airfare analysis supporting the advance-planning policy, and an **automatic exception report** (late booking, hotel/flight above grade, over-budget claims, missing or duplicate bills, and more).

The app is a full **Progressive Web App (PWA)**: Chrome and Edge show "Install App" on desktop, Android shows "Add to Home screen", iPhone users get guided instructions, and an **Install App button** sits in the top menu bar. It works on desktop, tablet and mobile.

Key facts

Item	Detail
Technology	Node.js 22+ backend (built-in HTTP + SQLite), vanilla HTML/CSS/JavaScript PWA front-end – zero third-party packages
Database	SQLite file data/abc_travel.db (11 tables), auto-created on first run; bills stored in uploads/
Users & roles	Admin + 10 standard roles in a drop-down (MD, Business Head, Sales Manager, Sales Executive, HR Head, Accountant, Travel Assistant, Finance Manager, Operations Manager, Employee) + custom roles
Workflow	9-step workflow with send-back / reject at every approval stage, notifications and full audit trail
How to run	Double-click Start_ABC_Travel_App.bat (Windows) – opens http://localhost:8080
Quality	43 automated API checks + full browser end-to-end run across all 7 roles – all passed; Chrome reports zero PWA installability errors

2. Requirement Coverage

Every requirement in the brief and how it is delivered:

#	Requirement	Delivered as
1	Travel expense app for "ABC Private Limited" (multinational)	Branded PWA; Domestic & International trips; multi-currency expense lines with exchange rates to INR
2	Pre-travel approval workflow, clearly identified	Visual 9-step workflow stepper on every request; role-based work queues; workflow diagram on dashboard and policy page
3	Backend database for users and task details	SQLite database with users, sessions, auth_logs, travel_requests, request_history, expense_items, attachments, notifications, settings, authority_matrix, feature boards
4	Login page with email id and password; sign-up option	Login / Sign Up tabs; salted scrypt password hashing; 7-day sessions; lock-out after 5 failed attempts
5	First user becomes Admin	First successful sign-up is automatically given Admin rights (logged as such)
6	Admin adds employees and their email ids	Employees screen: add / edit / deactivate, bulk CSV import, reset password, grant admin
7	Employee signs up with the id entered by Admin; others refused	Sign-up checks the employee master; any other email id is refused and the attempt is logged
8	Employee can see and edit pending approvals	My Trips & Claims with filters; edit while Draft / Sent back / pending first approval; cancel before booking; approvers edit budgets and claim line amounts in their queue
9	Only Admin sees the entire employee dashboard	Analytics, All Requests, Employees, Settings and Logs are Admin-only (server-enforced)
10	Save all sign-in and sign-up details in database	auth_logs table: SIGNUP_SUCCESS / REJECTED, LOGIN_SUCCESS / FAILED / BLOCKED, LOGOUT, password and employee changes with IP & device; Admin log screen
11	Request: cities, days, approximate amount, flight, hotel, food & misc, purpose	All fields plus local conveyance, client, cost center, preferences, advance; auto days and total; auto-fill from policy
12	Business Head → Managing Director approval	Routed to employee's reporting BH, then MD; approve (revise budget) / send back / reject
13	Support person books flight & hotel as per Matrix of Authority (5/4/3 Star)	Travel Assistant booking desk shows entitlement; bookings above grade need justification and are flagged; e-ticket / voucher upload
14	Expense statement with bills attached after travel	Expense lines with category, vendor, bill no., currency; PDF/JPG/PNG bills (camera on phone); bills mandatory above limit
15	Claim → Business Head → HR Head (final)	Line-wise approved amounts with notes; send back loop to employee
16	Accountant: accounting and payment processing	Voucher no., GL code, cost center, payment mode, UTR, date; advance adjustment; net payable / recoverable
17	Plan travel well in advance for cheaper tickets	Configurable minimum advance days (default 14); live indicator; justification for late requests; late-booking exception; lead-time vs fare chart
18	Visual dashboard: employee, destination, airline wise; analytics; flag exceptions	Admin analytics with 8 charts, KPI tiles, date filters, exception report, CSV export
19	Roles drop-down for Admin	Role drop-down in employee form; list editable in Settings
20	Typical corporate features + option to add more	Notifications, audit trail, advances, multi-currency, print, backup, policy page, settings & feature switches, editable Matrix of Authority, feature suggestion board, enhancement prompts

#	Requirement	Delivered as
21	Full PWA install support + Install button in menu bar	Manifest, icons (incl. maskable), service worker, shortcuts, screenshots, beforeinstallprompt handling, iOS guidance, Install App button
22	Single ZIP with summary, prompts, examples, executables, supporting docs	ABC_Travel_Expense_Project.zip – see Section 13

3. Solution Architecture

The solution is intentionally simple to install and maintain – one folder, one program, one database file.

Layer	Implementation
Client (PWA)	public/index.html, app.js (single-page app, hash routing), charts.js (SVG charts), styles.css (responsive, light/dark), manifest.webmanifest, sw.js (service worker), icons
Server	server.js – Node.js HTTP(S) server, REST API (~45 endpoints), role checks, workflow engine, exception engine, file upload handling, CSV export, backup
Database	SQLite via Node.js built-in node:sqlite – WAL mode, foreign keys, indexes; no database server to install
File storage	uploads/ folder – bills, e-tickets, hotel vouchers, payment proofs (PDF/JPG/PNG/WEBP/HEIC, ≤5 MB, SHA-256 duplicate detection)
Launchers	launch.js (version check, flags, opens browser, --demo mode); Start_ABC_Travel_App.bat, Start_Demo_Mode.bat, Start_ABC_Travel_App.command, start_mac_linux.sh
Security	bcrypt password hashes, random 256-bit session tokens, login throttling, server-side role enforcement on every endpoint, no self-approval, upload type/size whitelist, full audit trail

4. User Roles and Access

Role	What the role does in the app
Admin	First user. Adds employees and email ids, assigns roles/Business Heads, resets passwords, sets policy & Matrix of Authority, views analytics, all requests, sign-in/sign-up log, exports and backups. Can act on any stage (recorded as Admin override).
Sales Executive / Sales Manager / any employee	Raises pre-travel requests, edits pending requests, cancels, receives bookings, submits expense statement with bills, tracks reimbursement.
Business Head	First approver of travel requests (can revise budget) and of expense statements (can edit approved amount per line). Own requests skip this stage.
Managing Director	Final pre-travel approval. Own requests go directly to booking.
Travel Assistant	Booking Desk: books flights and hotels as per the Matrix of Authority and attaches tickets / vouchers.
HR Head	Final approval of the expense statement.
Accountant	Accounts Desk: disburses travel advances, posts vouchers, adjusts advances and records payment.
Finance Manager, Operations Manager, custom roles	Behave as employees with their own entitlement row in the Matrix of Authority.

Rules: nobody can approve their own request; a request waiting for a specific Business Head is visible only to that Business Head; every action and comment is stored in the request history.

5. Travel Workflow

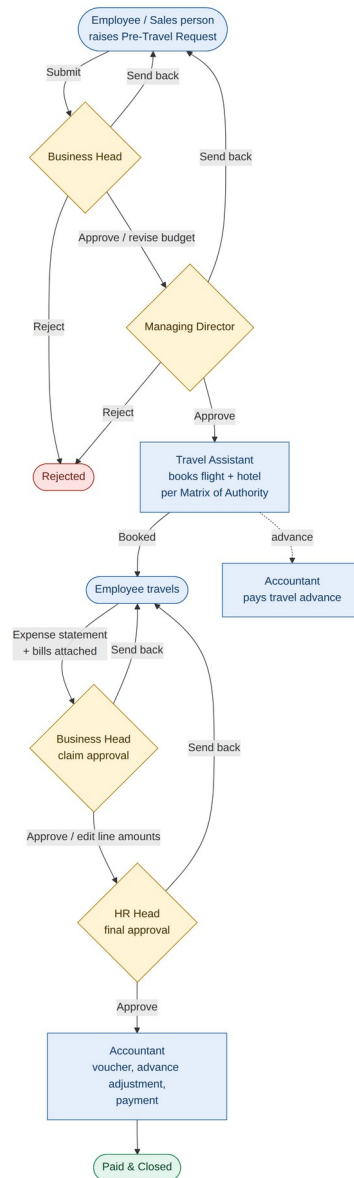


Figure 1 – Pre-travel approval, booking, expense statement and payment workflow

Stage	Owner	Actions available
Draft / Sent back	Employee	Edit, submit, cancel
Pending Business Head	Reporting Business Head	Approve (revise budget), send back, reject; employee can still edit until first action
Pending Managing Director	Managing Director	Approve, send back, reject
Pending Travel Booking	Travel Assistant	Book flight + hotel (justify if above entitlement), attach tickets, send back; Accountant may pay advance
Booked	Employee	Travel; add expense lines and bills; submit expense statement
Claim Pending Business Head	Business Head	Edit approved amount per line, approve, send back, reject
Claim Pending HR Head	HR Head	Final approval, send back, reject
Pending Accounting & Payment	Accountant	Voucher, GL code, cost center, payment mode/UTR/date; advance adjustment

Stage	Owner	Actions available
Paid & Closed	–	Read-only; printable statement

6. Matrix of Authority and Travel Policy

Default travel entitlements (editable by Admin in Policy & Settings). They drive the request form hints, the Travel Assistant's booking limits, the auto-fill estimates and the exception engine.

Role	Flight	Hotel	Hotel max / night	Food DA / day	Local / day
Managing Director	Business	5 Star	₹15,000	₹3,000	₹2,000
Business Head	Premium Economy	5 Star	₹10,000	₹2,500	₹1,500
HR Head	Economy	4 Star	₹7,000	₹2,000	₹1,200
Finance Manager	Economy	4 Star	₹7,000	₹2,000	₹1,200
Operations Manager	Economy	4 Star	₹6,500	₹1,800	₹1,000
Sales Manager	Economy	4 Star	₹6,000	₹1,500	₹1,000
Admin	Economy	4 Star	₹6,000	₹1,500	₹1,000
Sales Executive	Economy	3 Star	₹4,000	₹1,000	₹800
Accountant	Economy	3 Star	₹4,000	₹1,000	₹800
Travel Assistant	Economy	3 Star	₹4,000	₹1,000	₹800
Employee	Economy	3 Star	₹3,500	₹900	₹600

Policy rules (configurable)

- **Advance planning:** request at least 14 days before departure; late requests need a justification and are flagged.
- **Bills:** mandatory for each expense above ₹500.
- **Over-budget tolerance:** claims more than 10% above the approved budget are flagged.
- **Claim deadline:** expense statement within 15 days of return; overdue claims are flagged.
- **Maximum trip length:** 30 days guideline.
- **Feature switches:** international travel & multi-currency, travel advance, feature board, mandatory late-booking justification.

7. Corporate Features Included

- **Role-based work queues:** Pending Approvals (BH / MD / HR), Booking Desk (Travel Assistant), Accounts Desk (Accountant) with count badges.
- **Notifications:** in-app bell for every submission, approval, send-back, booking, advance and payment.
- **Audit trail:** every create, edit, approval, budget revision, booking, advance and payment with user, role, time and comment.
- **Travel advance:** requested in the form, paid by the Accountant, automatically adjusted at settlement (net payable or recoverable).
- **Bill attachments:** per expense line and per request; camera capture on phones; duplicate file detection.
- **Multi-currency:** foreign-currency expenses converted to INR at editable default rates.
- **Printing:** print-friendly travel & expense statement.
- **Exports & backup:** full register as CSV (Excel), one-click database backup.

- **Employee master:** CSV bulk import, deactivate leavers, reset passwords, reporting Business Head mapping.
- **Travel policy page** showing rules, workflow and Matrix of Authority to all employees.
- **Extensibility:** editable roles, departments, expense categories, airlines, currencies; feature suggestion board with voting and status; ready-made prompts for 14 future enhancements.

8. Analytics Dashboard and Exception Flags

Available to Admin only, filterable by travel date (custom, financial year, last 90 days, all time).

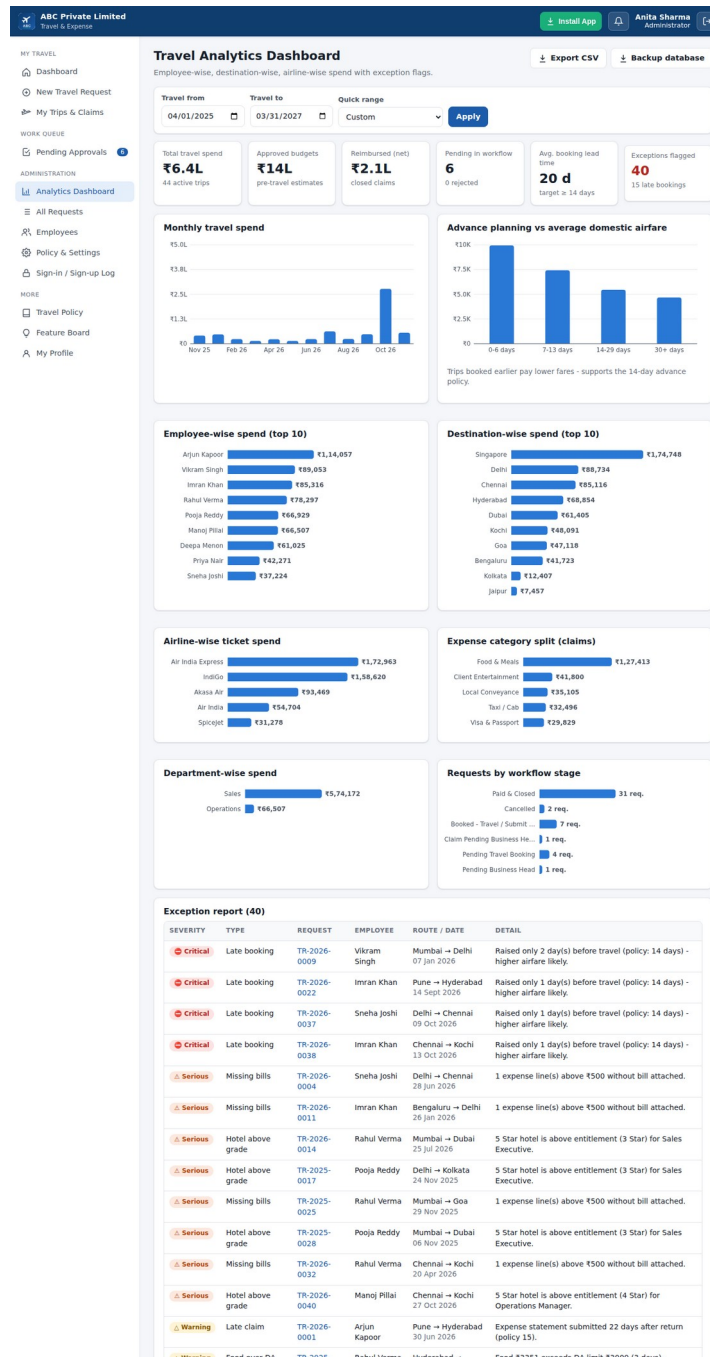


Figure 2 – Analytics dashboard with demo data

- KPI tiles: total spend, approved budgets, net reimbursed, pending in workflow, average booking lead time, exceptions.
- Charts: monthly spend; advance planning vs average domestic airfare; employee-wise; destination-wise; airline-wise; expense category; department-wise; workflow stage.
- Exception report and employee-wise summary table; each row opens the request.

Exception flags

Flag	Severity	Raised when
Late booking	Warning / Critical	Request raised fewer than the minimum advance days before travel (critical under 3 days)
Hotel above grade	Serious	Hotel category higher than the role's entitlement
Hotel rate over limit	Warning	Nightly rate above the entitlement limit
Flight class above grade	Serious	Flight class higher than entitlement
Over budget	Serious	Claim exceeds approved budget by more than the tolerance
Missing bills	Serious	Expense line above the bill limit without a bill
Duplicate bill	Critical	Same category, amount, date and bill no. repeated; same file uploaded before
Food over DA	Warning	Food total above daily allowance × days
Date outside trip	Warning	Expense dated outside the travel period
Long trip	Warning	Trip longer than the maximum days guideline
Late claim / Claim overdue	Warning	Expense statement submitted, or still pending, after the deadline

9. Progressive Web App (Install Support)

- **Web manifest:** name, short name, standalone display, theme colour, start URL and scope, 192/512 px icons plus maskable icon, three app shortcuts, wide and narrow screenshots for the richer install dialog.
- **Service worker:** caches the app shell for fast start and offline opening; API calls always go to the server so data is live.
- **Install App button** in the top menu bar (and on the login page) triggers the browser's install prompt; on iPhone/iPad or unsupported browsers it shows step-by-step instructions.
- **Verified:** Chrome DevTools reports no installability or manifest errors.

Important: browsers allow installation only from <https://> or <http://localhost>. Use HTTPS for phones and other PCs (see [Deployment_and_HTTPS_Guide.md](#)).

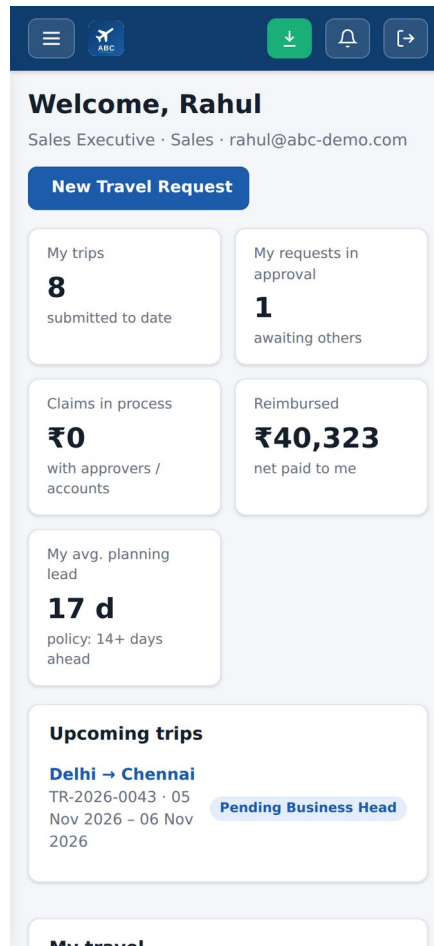


Figure 3 – Mobile view with Install button (green) in the menu bar

10. Database Design

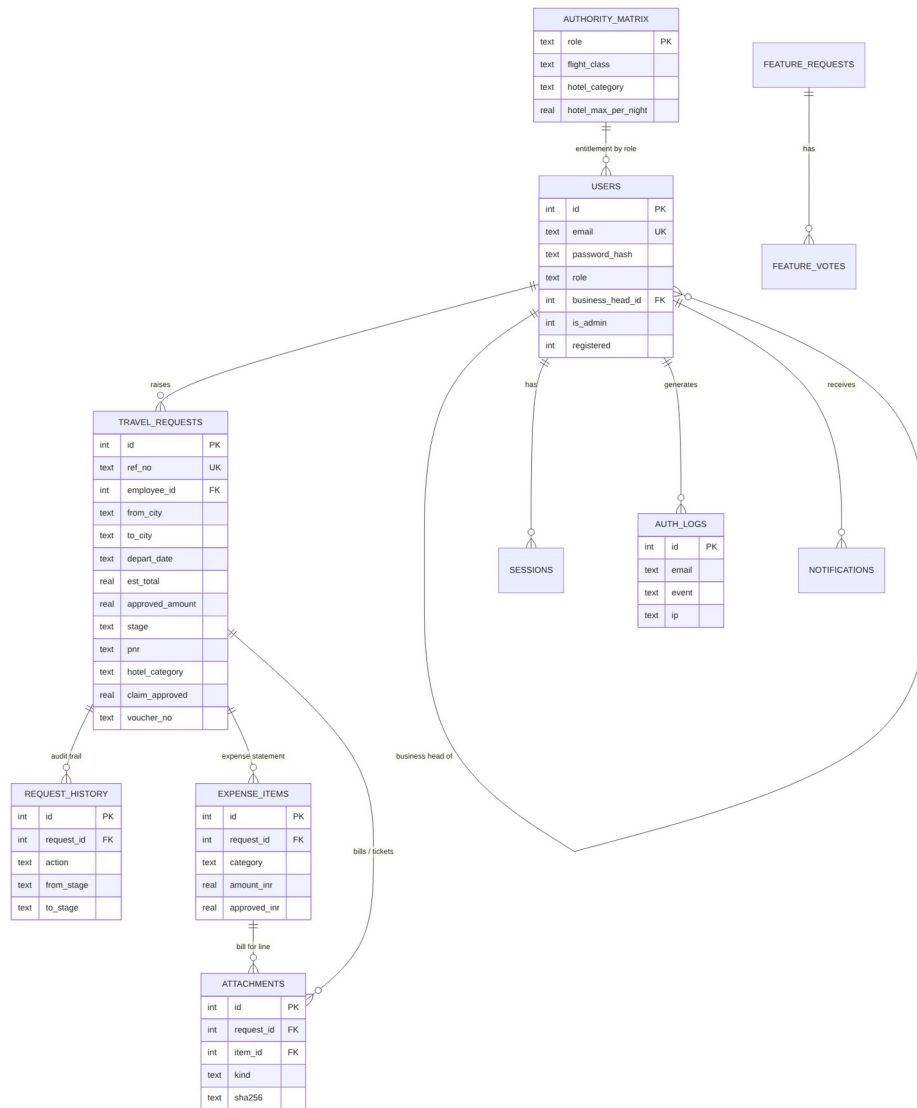


Figure 4 – Entity relationship diagram (key columns)

Table	Purpose
users	Employee master and login credentials (hash + salt), role, department, reporting Business Head, admin flag, sign-up status
sessions	Active login sessions with expiry, IP and device
auth_logs	All sign-up, sign-in, failed, blocked, logout, password and employee/settings change events
travel_requests	Request, estimates, approval, booking, claim totals and payment details, current stage
request_history	Audit trail of every workflow action with comments
expense_items	Expense statement lines, currency, INR amount, approved amount and approver note
attachments	Bills, tickets, vouchers, payment proofs (metadata + SHA-256)
notifications	In-app notifications per user
settings / authority_matrix	Policy values, master lists, feature switches / role entitlements
feature_requests / feature_votes	Feature suggestion board

Full DDL: 05_Supporting_Documents/Database_Schema.sql.

11. Installation and Getting Started

Requirements

Windows 10/11 (or macOS / Linux) with Node.js 22 LTS or newer. The Windows launcher installs Node.js automatically via winget if it is missing. No other software or database server is needed.

Start the app

1. Unzip ABC_Travel_Expense_Project.zip.
2. Open 04_Executable_Application\ABC_Travel_Expense_App and double-click **Start_ABC_Travel_App.bat**.
3. The browser opens at <http://localhost:8080>. Keep the console window open (it is the server).
4. Click **Sign Up** – the first person becomes Administrator. Log in.
5. Go to **Employees** and add Business Head, Managing Director, Travel Assistant, HR Head, Accountant and sales staff (or import Sample_Employee_Import.csv).
6. Each employee opens the same address, signs up with the email id the Admin entered, and logs in.
7. Click **Install App** in the menu bar to install it as a desktop / mobile app.

Try it with demo data

Double-click **Start_Demo_Mode.bat** – it builds a demo database (15 employees, 46 trips at every stage with exceptions) and opens <http://localhost:8081>. Log in as admin@abc-demo.com / Demo@1234 (all demo users use Demo@1234).

Day-to-day use by role

User	Steps
Employee	New Travel Request → fill route, dates, purpose, estimates (or Auto-fill) → Submit. After booking: open the trip → Add expense + attach bill for each item → Submit expense statement.
Business Head	Pending Approvals → open → revise budget if needed → Approve / Send back / Reject. For claims: adjust line amounts with notes → Approve.
Managing Director	Pending Approvals → open → Approve / Send back / Reject.
Travel Assistant	Booking Desk → open → enter airline, flight, PNR, fare, hotel, category, nights, rate → attach e-ticket → Confirm booking.
HR Head	Pending Approvals → review claim and bills → Approve.
Accountant	Accounts Desk → record advances; for approved claims enter voucher, GL, payment mode, UTR, date → Post voucher & mark paid.
Admin	Analytics Dashboard, All Requests (Export CSV), Employees, Policy & Settings, Sign-in / Sign-up Log, Backup database.

12. Testing and Quality

Test	Scope	Result
Automated API workflow test (npm test)	Sign-up rules, admin, employees, request, edit rules, BH/MD approvals, advance, Matrix of Authority, bills, claim approvals with send-back, payment and advance adjustment, special routing, analytics access, logs, notifications	43 / 43 passed
Browser end-to-end test	Complete trip through the real UI with 7 different logins on a fresh database	Passed – Paid & Closed
PWA audit	Chrome installability & manifest checks, service worker	No errors
Responsive check	Phone 390 px, 540 px, laptop 1280 / 1366 px	Passed

Test	Scope	Result
Data tools	CSV import (8 employees), CSV export, demo data generator	Passed

Detailed log: 05_Supporting_Documents/Test_Report.txt.

13. Contents of the ZIP File

Folder / file	Description
00_READ_ME_FIRST.txt	Quick start and index
01_Summary_Document/ ABC_Travel_Expense_Project_Summary.docx (.pdf)	This document
02_Prompt_Files/01_Original_User_Prompt.txt	The original requirement as given
02_Prompt_Files/ 02_Detailed_Build_Specification_Prompt.md	Reusable detailed specification prompt
02_Prompt_Files/ 03_Future_Enhancement_Prompts.md	14 ready prompts for adding more features
03_Example_Files/Sample_Bills/	Sample hotel invoice, taxi receipt, restaurant bill, airline e-ticket (PDF, marked as samples)
03_Example_Files/ Sample_Employee_Import.csv	Template for bulk employee import
03_Example_Files/ Example_Demo_Database_abc_travel_demo.db	Demo SQLite database with 46 trips
03_Example_Files/ Example_Travel_Register_Export.csv	Example CSV export from the demo
03_Example_Files/ Example_Walkthrough_Test_Scenario.md	Step-by-step demo / training script
03_Example_Files/Example_Screenshots/	26 screenshots of every screen and role, desktop and mobile
04_Executable_Application/ ABC_Travel_Expense_App/	Complete application: launchers (.bat / .command / .sh), server.js, launch.js, public/ (PWA), tools/, tests/, README.md
05_Supporting_Documents/	Workflow diagram, ER diagram, Database_Schema.sql, API_Reference.md, Matrix_of_Authority_Default.csv, PWA_Installation_Guide.md, Deployment_and_HTTPS_Guide.md, Test_Report.txt

14. Notes, Limitations and Roadmap

- **Runtime:** the "executable" is a one-click launcher that runs on Node.js 22+ (installed automatically on Windows). A self-contained .exe was not produced because it requires platform binaries that were not available in the build environment.
- **Notifications** are in-app; email / WhatsApp / push alerts are listed as enhancements.
- **Mobile installation** requires the app to be served over HTTPS (browser rule).
- **Fares** are entered by the Travel Assistant; live airline / travel-agency integration is a roadmap item.
- **Exchange rates** are maintained by Admin in Settings.

Suggested next features: email alerts, delegation of authority, SLA escalation, OCR of bills, corporate card reconciliation, Tally Prime voucher export, department budgets, single sign-on, web push, multi-entity support (see 03_Future_Enhancement_Prompts.md).